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EVEN END SEMESTER EXAMINATION - 2025

Name of the Program: B.Tech Aerospace Engineering *Semester IV*

Name of the Course: Aircraft Performance *Course Code:* **TAS 404**

Time: 3 Hours Maximum Marks: 100

Note:

(i) All questions are compulsory.

(ii) Answer any two sub questions among a, b & c in each main question.

(iii) Total marks in each main question are twenty.

(iv) Each question carries 10 marks

Q1	**(20 marks)**	
(a)	Develop equations for the pressure and density as a function of altitude from 0-21 KM. The analytical model of the atmosphere should make use of the hydrostatic equation, for which the density is eliminated through the use of the equation of state for a thermally perfect gas. Assume that the temperature of the air from 0-11 KM is given by $T = T_o - \lambda h$.	
(b)	An airplane has a wing of planform area 51.22 m^2 and span 20 m. It has a fuselage of frontal area 3.72 m^2 and two nacelles having a total frontal area of 3.25 m^2. The total planform area of horizontal and vertical tails is 18.6 m^2 . Obtain a rough estimate of the drag polar in a flight at a speed of 430 kmph at sea level (S.l.). when landing gear is in retracted position. Assume CD π for Wing=0.0054, fuselage=0.080, Nacelles=0.1, Tail Surface=0.006 & Adding 10% for interference effects.	CO4
(c)	Consider a flying wing with a wing area of 206 m^2, an aspect ratio of 10, a span effectiveness factor of 1.0, and a NACA 4412 airfoil. The weight of the airplane is 7.5 x 10^5 N, and the flight velocity is 100 m/sec. calculate the total drag on the aircraft at sea level. Assume L = W and the profile drag coefficient is 0.006.	
Q2	**(20 marks)**	
(a)	What is boundary layer control? Explain with suitable sketches, different method of control of the behavior of the boundary layer.	
(b)	Why Flaps are lowered during landing and take-off of an Airplane.	CO1
(c)	What is the effect of Altitude on the Maximum and minimum Speed of an Aircraft.	
Q3	**(20 marks)**	
(a)	Derive the General relation between the Total drag Vs Velocity, Show this with graphical representation. Also discuss which drag will be more in case of high speed aircraft and what factor to be considered to reduce .	CO3
(b)	Prove that the excess power during the takeoff climb is equal to sum of the time rate of the potential energy and kinetic energy.	

(c)	For a minimum power show that the zero lift drag has to be one third of the lift dependent drag	
Q4	**(20 marks)**	**CO5**
(a)	A sail plane weighs 4500 N and has a wing loading of 600 N m^{-2} . Its drag equation is $$C_D = 0.010 + 0.022\, C_L^2$$ After completing launch at 350 m in still air, what is the greatest distance the sailplane can cover, and what is the greatest duration of flight possible, assuming in both cases flight over level ground? Find also the corresponding speeds of flight. Ignore changes of density of the atmosphere	
(b)	For a Subsonic Aircraft show that the rate of climb is given by $$\frac{dh}{dt} = \frac{(T-D)V}{W(1+\frac{VdV}{gdh})}$$	
(c)	An aircraft weighing 250 000 N has a wing area of 90 m^2, and its drag equation is $$C_D = 0.016 + 0.04\, C_L^2$$ Calculate the minimum thrust required for straight and level flight, and the corresponding true air speeds at sea level. Calculate also the minimum power required, and the corresponding true air speeds at sea level	
Q5	**(20 marks)**	**CO2**
(a)	Describe Aircraft *Take-off* and *Landing* procedures. Derive the relations for runway distance required for any one of the cases.	
(b)	Determine the expression for maximum turn rate as a function of wing loading , load factor and maximum lift.	
(c)	Find the correct angle of bank of an airplane turning on a circle of radius 120m at a velocity of 200 Km/h. Also compare the lift on the airplane to the lift during a cruising flight	